

Considering only variability in annual recruitments in the sardine assessment

C.L. de Moor*

Correspondence email: carryn.demoor@uct.ac.za

Following requests by the Technical Task Group of the Small Pelagic Scientific Working Group, the impact of variability in recruitment parameters only is considered as an intermediate step to the hybrid approach for MCMC.

Keywords: assessment, posterior distributions, recruitment, sardine, South Africa

Introduction

de Moor (2025a) assumed variability in most model parameters for the Operating Models for the Scoping Management Strategy Evaluation. Following requests by the Technical Task Group of the Small Pelagic Scientific Working Group, the impact of variability in recruitment parameters only is considered in this document as an intermediate step.

Methods

The models used differ from those used by de Moor (2025a), and assume fixed values of $M_{w,y,0}^{CTS} = 1.0$, $M_{w,y,1+}^{CTS} = 0.8$, $M_{s,y,0}^{WTS} = 1.25$ and $M_{s,y,1+}^{WTS} = 1.25$ (de Moor 2025b). S_0 assumes no stock recruitment relationship, and historical October recruitments are estimated independently (de Moor 2025c). S_{BH} assumes a Beverton Holt stock recruitment relationship with $\sigma_{r,j}^2 = 5$ (de Moor 2025d).

For the Markov Chain Monte Carlo runs, most parameters were fixed at the joint posterior mode resulting from a given set of starting values. Jittering has not been undertaken to ensure the set of parameters from which the chains begin corresponds as close as possible to the joint posterior mode. However, a test using a set of substantially different initial parameters values was run. For S_0 , posterior distributions for $R_y^{CTS}/10 \sim U(0,50)$ and $R_{c,y}^{WTS}/10 \sim U(0,50)$, $y_0 \leq y \leq y_n$ were required, while for S_{BH} , posterior distributions for $\varepsilon_y^j \sim N(0, \sigma_{r,j}^2)$, $y_1 \leq y \leq y_n$, with $-20 \leq \varepsilon_y^j \leq 20$ were required.

Given time constraints, full convergence diagnostics have not yet been considered.

Results and Discussion

The S_0 models converged on very similar parameter values, even when begun from rather different initial parameter values. However, the S_{BH} models converged on very different parameter values for K^{WTS} , with the same objective function to the 6th decimal place (Figure 1). There was also some difference (e.g., 0.320, 0.321, 0.343) in h^{WTS} . The three years of pulse recruitment are not included in the Beverton Holt stock recruitment relationship for WTS.

For S_0 , two chains of approximately 5000 000 were thinned by 1000, and 1000 sets of parameters were resampled with replacement after a burn-in of 1000. Figure 2 shows the historically estimated and future projected recruitment under a no future catch scenario, according to the Operating Model of de Moor (2025e,f).

* MARAM (Marine Resource Assessment and Management Group), Department of Mathematics and Applied Mathematics, University of Cape Town, Rondebosch, 7701, South Africa.

Initial chains of S_{BH} indicated slow mixing for some residuals once largely negative residuals were selected. Positive samples of ε_y^j never increased above 4. A new set of chains with $-5 \leq \varepsilon_y^j \leq 5$ are currently being explored. In addition, alternative prior distributions on K^j are also being explored (e.g. Figure 3).

References

- de Moor CL. 2025a. Posterior distributions of key model parameters from the sardine assessments to be used as input for the Operating Models for the Scoping MSE. DFFE: Branch Fisheries Document FISHERIES/2025/JUN/SWG-PEL/31.
- de Moor CL. 2025b. Exploring further key uncertainties about the sardine assessment. DFFE: Branch Fisheries Document FISHERIES/2025/SEP/SWG-PEL/49.
- de Moor CL. 2025c. The updated assessment model for the revised sardine stock structure hypothesis. DFFE: Branch Fisheries Document FISHERIES/2025/APR/SWG-PEL/17rev.
- de Moor CL. 2025d. Investigating the impact on the South African sardine assessment of estimating Beverton Holt stock recruitment relationships during conditioning. DFFE: Branch Fisheries Document FISHERIES/2025/MAY/SWG-PEL/28.
- de Moor CL. 2025e. Initial testing of the Scoping MSE for South African sardine. DFFE: Branch Fisheries Document FISHERIES/2025/JUN/SWG-PEL/32.
- de Moor CL. 2025f. Comparing an initial set of candidate Management Procedures for the Scoping MP for South African sardine. DFFE: Branch Fisheries Document FISHERIES/2025/JUL/SWG-PEL/41.

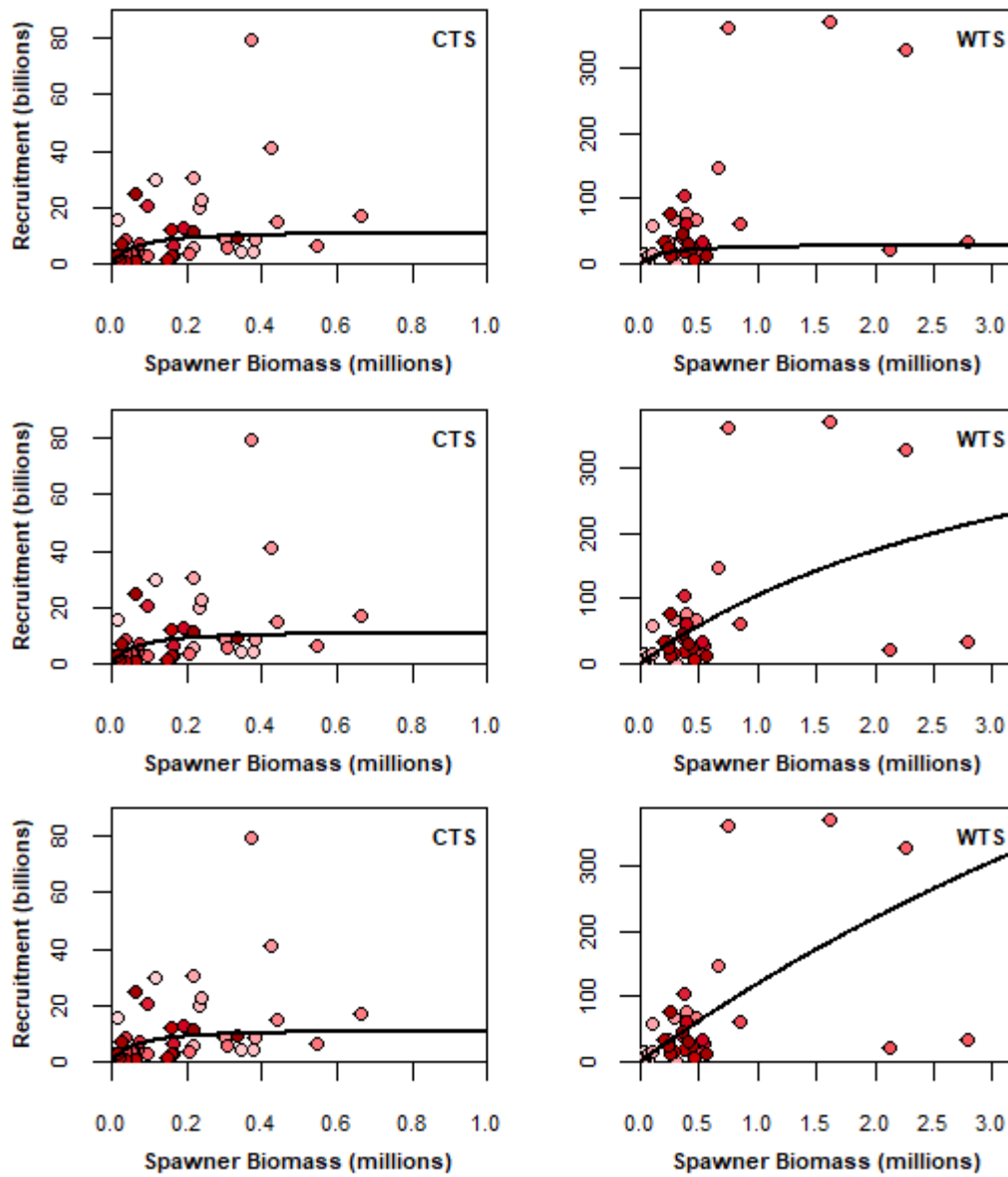


Figure 1. The estimated spawner biomass and October recruitment for CTS (left) and WTS (right) for S_{BH} which results from three different sets of initial parameter values.

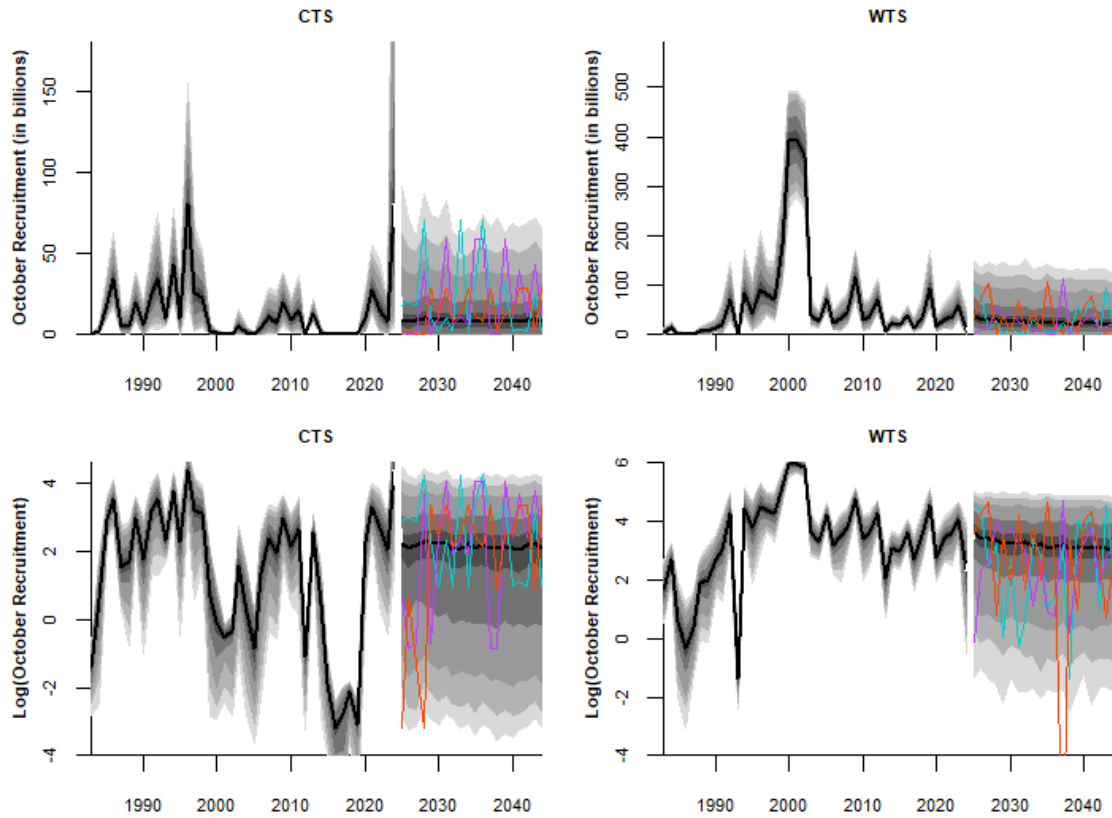


Figure 2a. Median (solid line), 20, 50, 80, 90 and 95% probability intervals (in increasingly lighter shades of grey) of historically estimated and projected October recruitment for S_0 under a no future catch scenario. Trajectories of three replicates are also shown in colour. The lower row is a repeat of the top, but with recruitment in log-space, in accordance with the error distribution assumed.

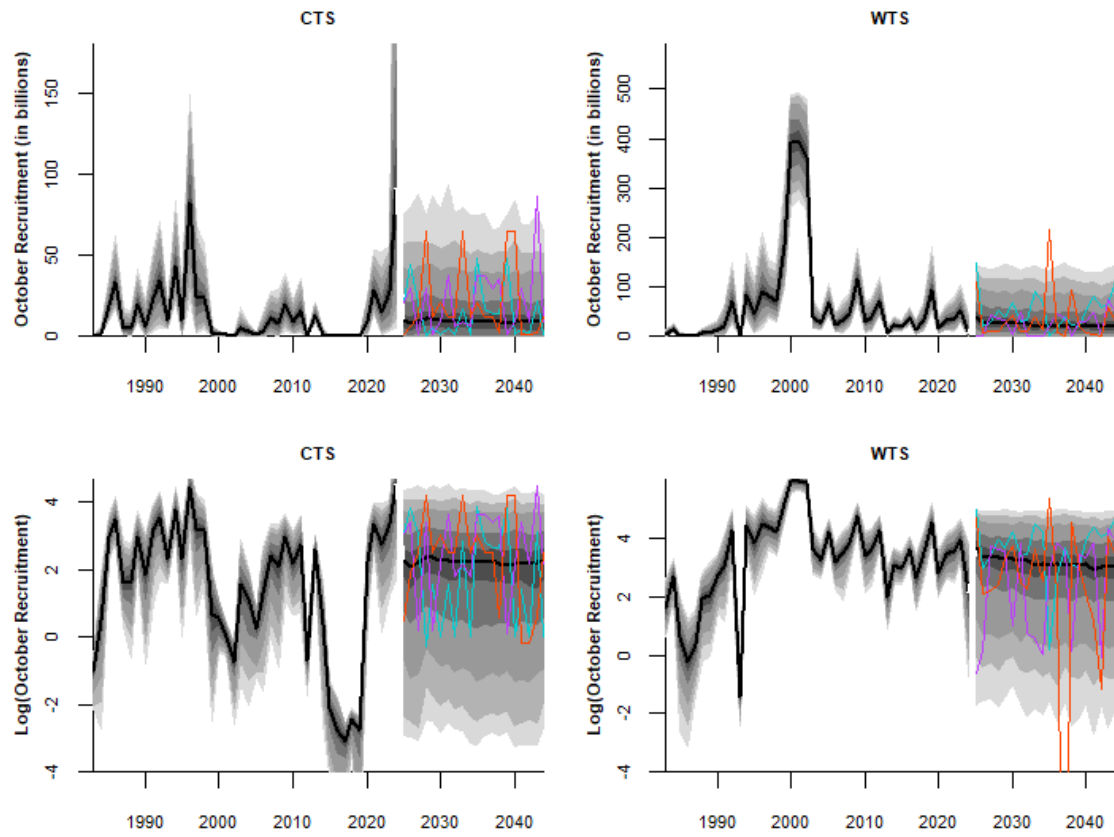


Figure 2b. As for Figure 2a, but from a different set of initial parameter values.

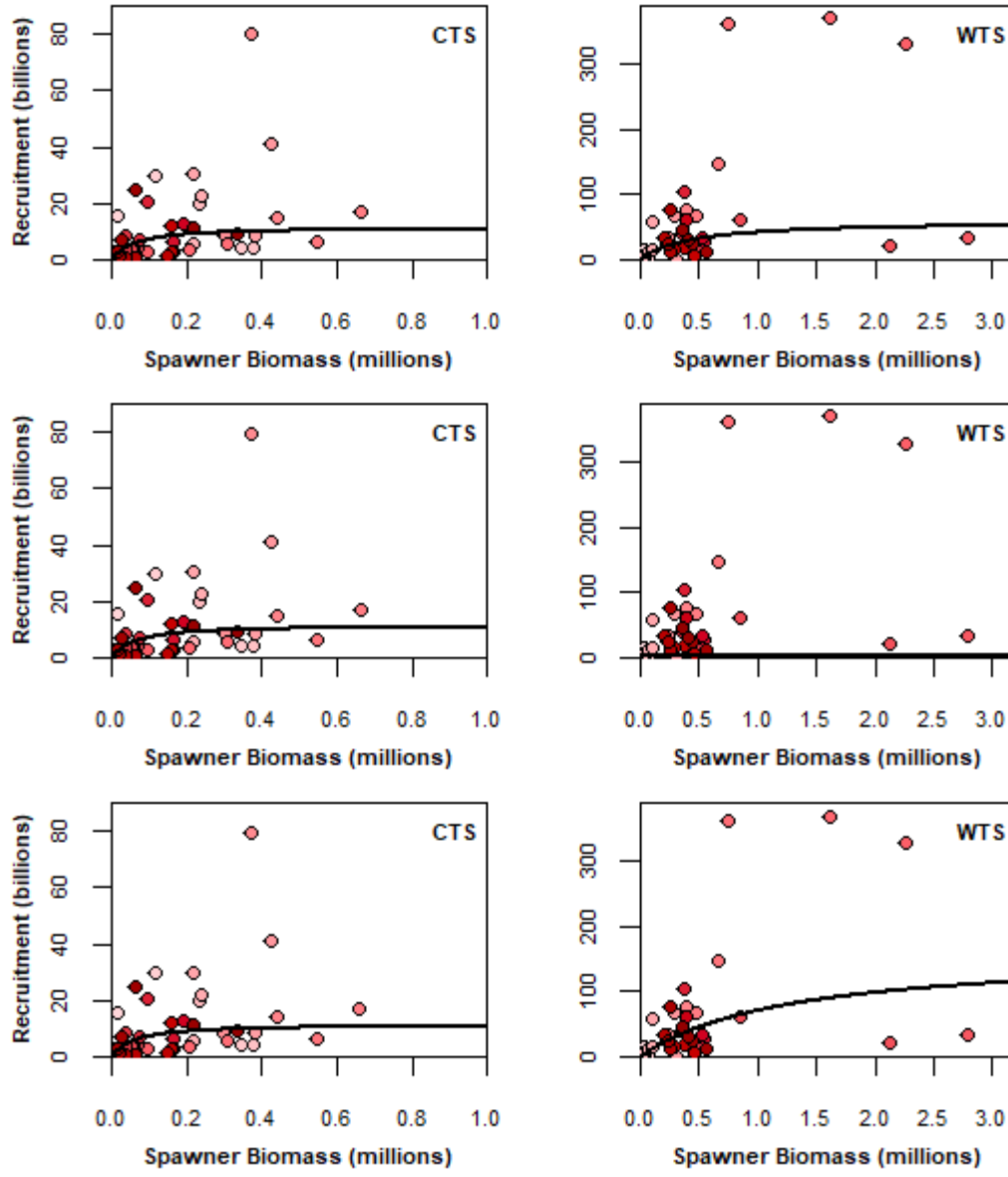


Figure 3. A repeat of Figure 1, but assuming $K^j/1000 \sim \left(1 + e^{(K^j/1000-5)/1}\right)^{-1}$.